FEDERAL SPECIFICATION

COMPONENTS FOR INTERIOR ALARM SYSTEMS

FOIL

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

The complete requirements for procuring the foil described herein shall consist of this document and the latest issue of W-A-450/GEN.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers Type II sensors, Foil, under Federal specification W-A-450.

1.2 Classification.

Size I - For use on plain glass.

Size II - For use on wired glass, walls, ceilings, floors and doors.

Size III - For use on walls, ceilings, floors and doors.

2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications.

W-A-450 Components for Interior Alarm Systems

(Activities outside the Federal Government may obtain copies of the Federal Specifications, Standards and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402).

(Single copies of the specification and other Federal

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Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in the agencies.)

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on the date of invitation for bids or request for proposal shall apply.

Underwriters' Laboratories, Inc. (UL) Publications

UL 606 Linings and Screens for Use With Burglar Alarm Systems

UL 681 Installation and Classification of Mercantile and Bank Burglar Alarm Systems

(Application for copies should be addressed to Underwriters' Laboratory, 333 Pfingsten Rd., Northbrook, IL 60062.)

3. REQUIREMENTS

3.1 Materials. Materials for components shall be as specified herein. Material not specified shall be the same as material used commercially and shall be suitable in all respects for the intended purpose.

3.1.1 Foil. Foil shall be consist of lead (60%), zinc (20%) and tin (20%).

3.2 Dimensions.

3.2.1 Size I. Size I foil shall be 9.5 to 12.7 mm wide and not more than 0.038 mm thick.

3.2.2 Size II. Size II foil shall be 9.5 to 12.7 mm wide and not more than 0.08 mm thick.

3.2.3 Size III. Size III foil shall be 12.7 to 25.4 mm wide and not more the 0.08 mm thick.

3.3 Tensile strength Foil shall not exceed 6130 kPa in tensile strength.

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3.4 Conductivity. Foil shall be capable of carrying a maximum electrical current of 60 milliamperes at 60 volts with a temperature rise of not more than 1°C.

3.5 Operating range. All equipment shall be capable of operating within the minimum temperature range indicated below:

0°C to 49°C and relative humidity of 85% at 25.5°C.

3.6 Installation guidelines. Installation guidelines shall be provided with the foil. The installation guidelines shall be as per UL 681 and UL 606.

3.7 Workmanship. The foil shall be manufactured and finished in such a manner as to meet specification requirements.

4. QUALITY ASSURANCE PROVISIONS

4.1 Material inspection. The supplier is responsible for insuring that components and materials used are manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or if none, in accordance with this specification.

4.2 Testing. The foil shall be tested in accordance with UL 681.

5. PREPARATION FOR DELIVERY.

5.1 Packaging, packing and marking. The foil shall be packaged, packed and marked in accordance with W-A-450C/GEN.

6. NOTES

6.1 Intended use. Foil is a thin metal conductor intended to be bonded to a protected surface so that the conductor will break when the protected surface is cut or broken. Intrusion detection systems constructed with components qualified under this specification are intended to notify protective personnel (e.g., response force) whose responsibility is the protection of areas requiring special security measures. Properly assembled systems are highly resistive to neutralization or compromise. Guidance in selecting a particular item is not intended and is beyond the scope of this document.

6.2 Application. The purpose of the foil is to provide a means for detecting an intrusion into a normally secured area when the intrusion is affected by a destructive penetration through a glass door, window, partition, and etc. The type of foil, with

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necessary dimensions (length and width), will depend upon the application intended and shall be in accordance with UL specifications referenced above.

6.3 Installation testing. Test the foil tape resistance with appropriate equipment. The resistance of foil loop should be measured and be low enough to allow proper operation of the associated alarm zone, to verify there are no breaks, hairline cuts etc., in the foil.

6.4 Ordering data. Purchaser should exercise and desired options permitted herein and include the following information in procurement documents:

(a) Title, symbol, and date of this specification.
(b) Type.
(c) Quantity required.
(d) Destination inspection required.
(e) Levels of packaging and packing required.
(f) Whether special marking required.

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